Xuanning Liu

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EDUCATION

University of California, Los Angeles (UCLA) Los Angeles, CA
Master of Computer Science (MSCS) Expected Jun 2027
GPA: 4.0; NLP, Big Data Analytics, Interactive Robotics, Graphics Artificial Life

University of California, Los Angeles (UCLA) Los Angeles, CA
BS in Computer Science, Minor in Neuroscience Sep 2021 - Mar 2025
- **GPA**: 3.86; UPE Honor Society, 8 Dean's Honors List; AI/ML, Algorithms, Computational Neuroscience

SKILLS

Languages: Python, C++, Java, JavaScript (React), SQL, MATLAB
Frameworks/Libraries: PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas
Tools: Git, Linux, Docker, Unity, MySQL, AWS, Jupyter Notebook, LaTeX

EXPERIENCE

Privacy and Security in LLM | Python Los Angeles, CA
Research Assistant May 2024 - Dec 2024
- Developed a tool to mask sensitive data before sending it to GPT, using synthetic data and fine-tuning
- Created keyword-based privacy definitions and context-aware masking methods
- Established privacy metrics and benchmarks to ensure secure data handling

Computational Imaging and Microscopy Lab | Python, PyTorch Los Angeles, CA
Research Assistant Sep 2024 - Mar 2025
- Built a self-supervised learning model in Python and PyTorch for hologram reconstruction, optimizing training with synthetic data and physics-consistency loss
- Developed a robust pipeline to handle unknown perturbations in hologram distances and wavelengths
- Eliminated the need for labeled data, enhancing model adaptability to unseen biological samples

PROJECTS

Robust Active Contour Models for Biomedical Image Segmentation | NumPy, SciPy, scikit-image
- Developed classical, adaptive, multiscale, and reinforcement-learning-guided active contour models for noisy biomedical image segmentation
- Designed a combined multiscale and spatially adaptive algorithm, achieving 0.736 IoU on fluorescence microscopy and 0.631 IoU on ultrasound images
- Built reproducible evaluation pipelines using IoU, Hausdorff distance, and mean contour distance, reaching up to 0.997 IoU on synthetic benchmarks

Multi-Agent Squirrel Foraging Simulation | Unity 6, C#, Unity ML-Agents, PPO, Python, TensorBoard
- Developed a Unity ML-Agents simulation where autonomous squirrels learn foraging, predator avoidance, safe-zone navigation, and energy management through reinforcement learning
- Engineered a 36-value observation space, hybrid action space, and multi-objective reward function; trained a 3-layer PPO policy for 3 million steps and deployed it through ONNX inference
- Built procedural environment systems and experiment instrumentation for terrain-aware spawning, obstacle interactions, episode metrics, checkpoints, and TensorBoard analysis

Gymstant Web App | JavaScript, React, Firebase
- Developed a full-stack web application with authentication, user management, and real-time data storage using Firebase
- Implemented social content sharing features, enabling users to create, manage, and control the visibility of posts
- Optimized data synchronization and state management to improve responsiveness, reliability, and user experience